Biochemistry Cume
September 11, 2015

1) (25 points) Show that K_m for an enzymatic reaction is equal to the substrate concentration at which the rate is half maximal.

2) (50 points) Salicylate inhibits the catalytic action of glutamate dehydrogenase. Use the following data to determine whether the inhibition is competitive or noncompetitive. The concentration of salicylate is held constant at 40 mM. Also, calculate Km for glutamate dehydrogenase.

Substrate concentration (mM)	1.5	2.0	3.0	4.0	8.0	16
mg Product/min (Without Salicylate)	0.21	0.25	0.28	0.33	0.44	0.40
mg Product/min (With Salicylate)	0.08	0.10	0.12	0.13	0.16	0.16

3) (25 points) Given the equation for competitive inhibition:

$$v = \frac{Vmax[S]}{aKm + [S]}$$

Where $a = 1 + [I]/K_i$ and represents a function of both the competitive inhibitor's concentration and its affinity for the enzyme active site.

A) Is it possible for the value of "*a*" to be less than 1? If so, explain what would happen if the value of "*a*" was less than 1?